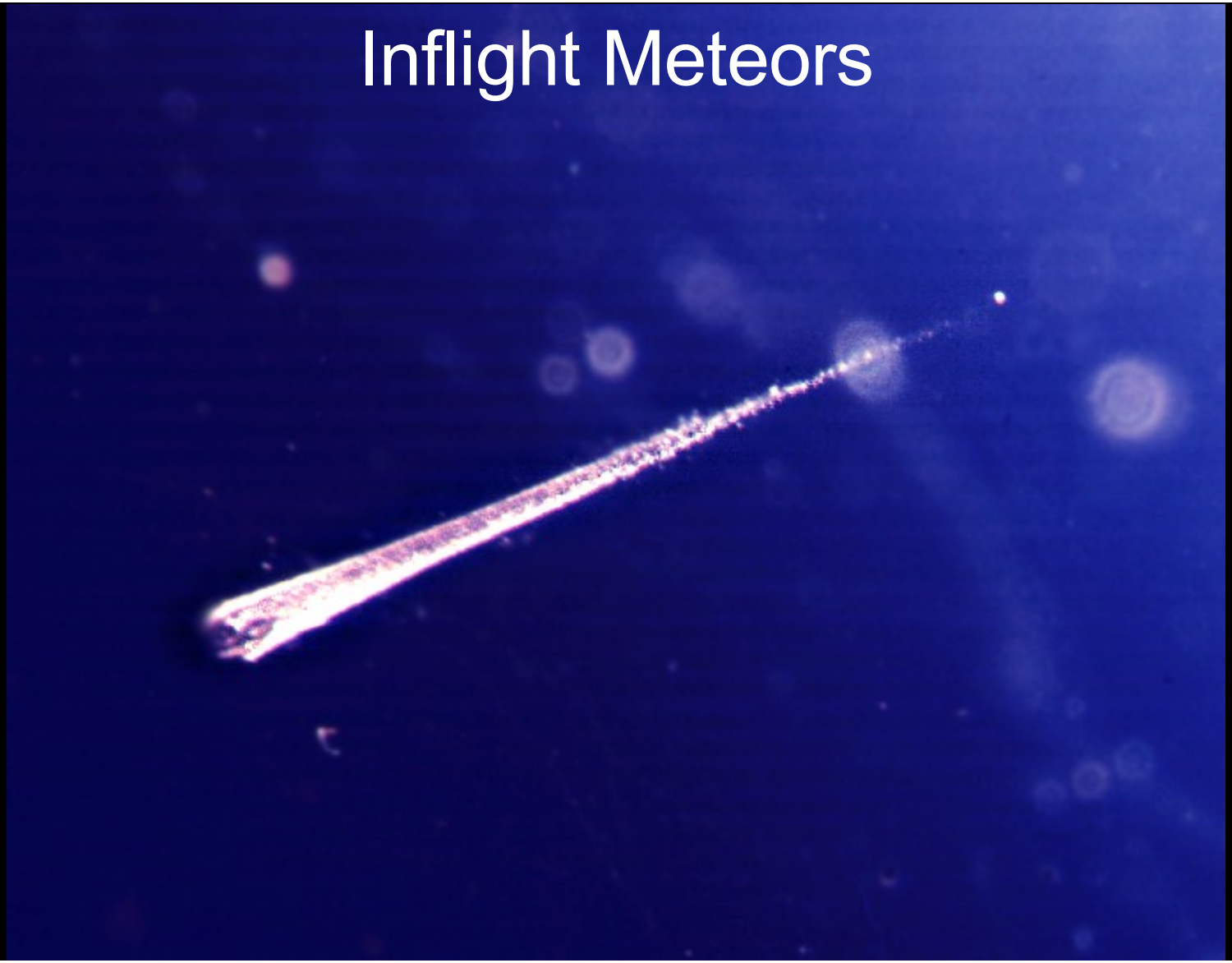
Introductory Radio Astronomy Projects

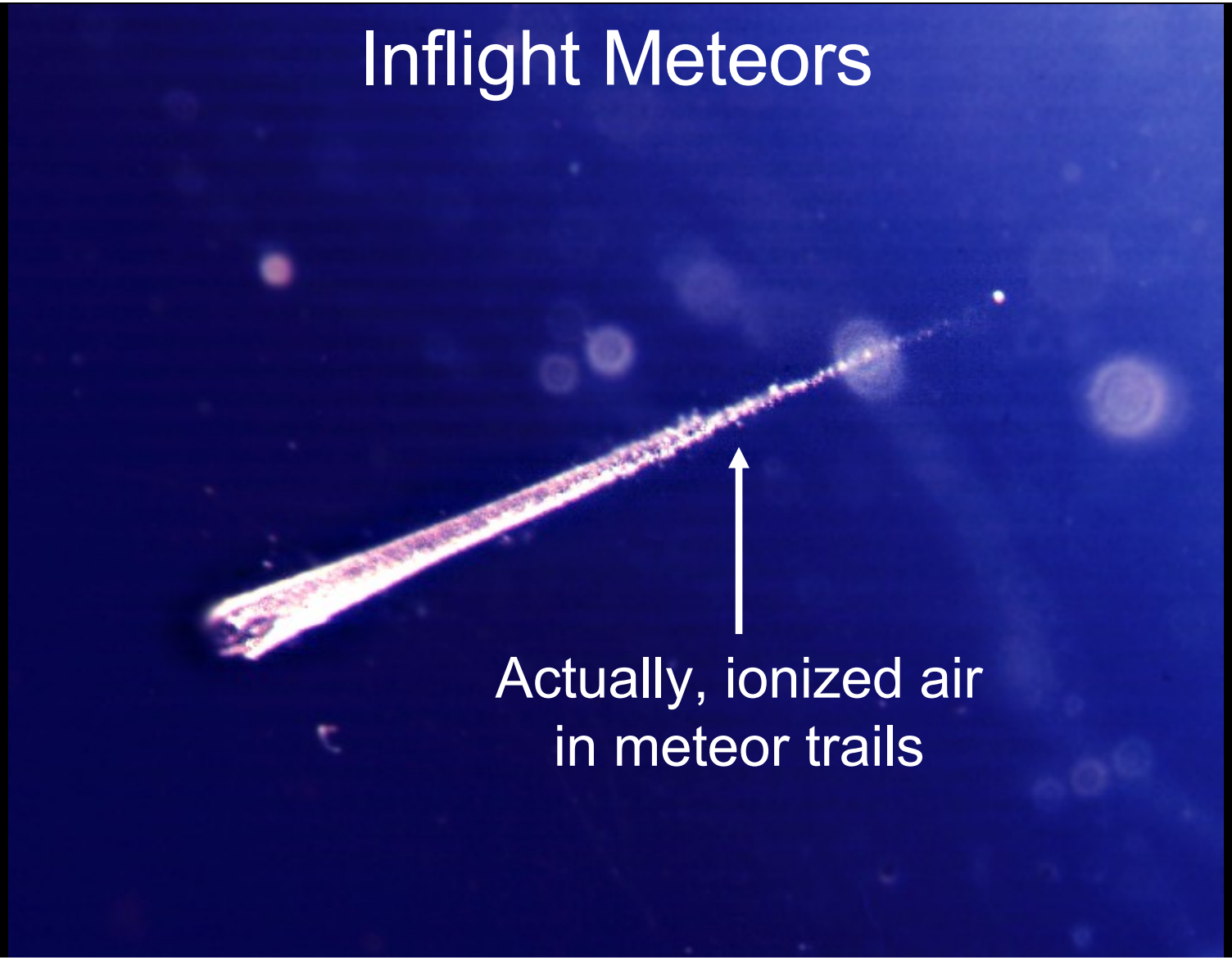
1. Itty Bitty Radio Telescope
2. Meteor Detection
3. SuperSID
4. Scope in a Box
5. Radio Jove



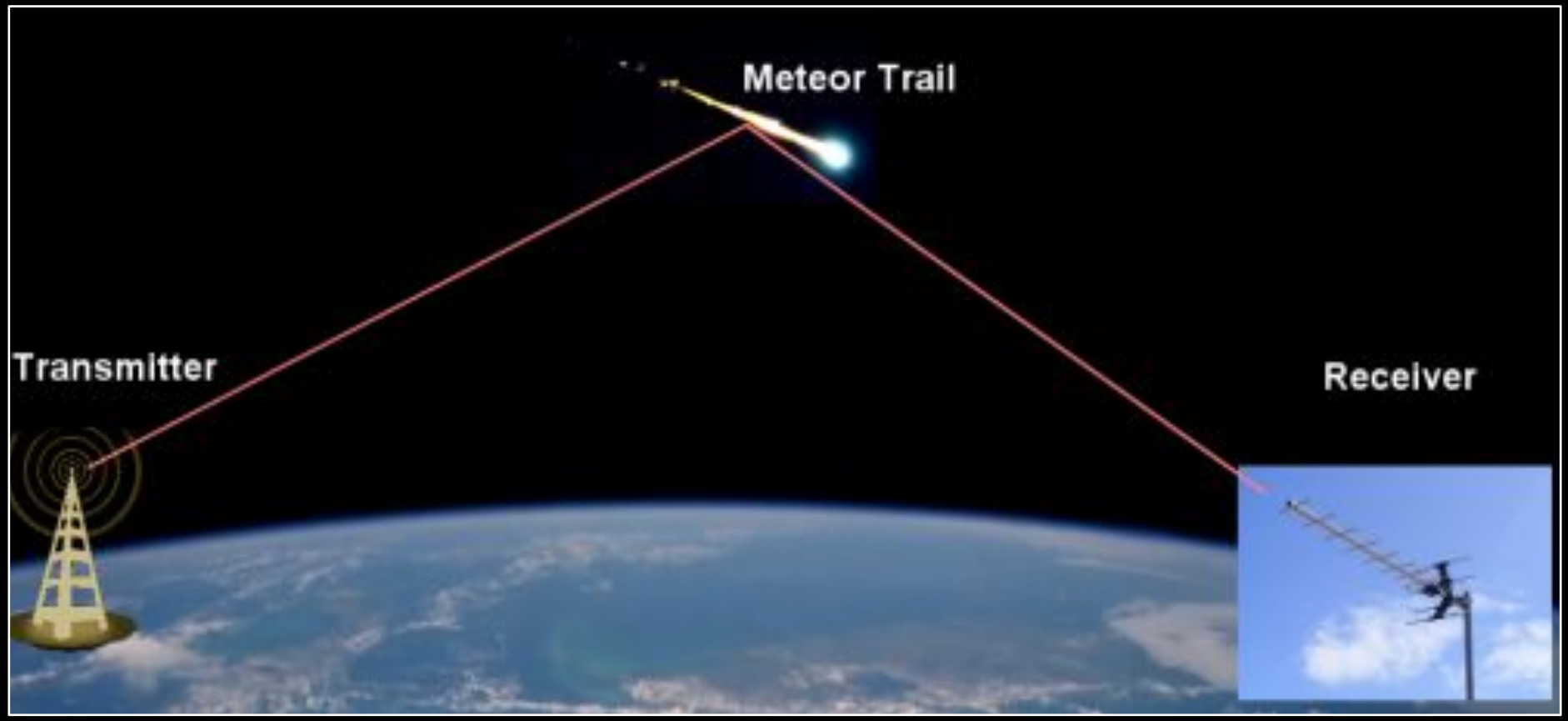
Inflight Meteors



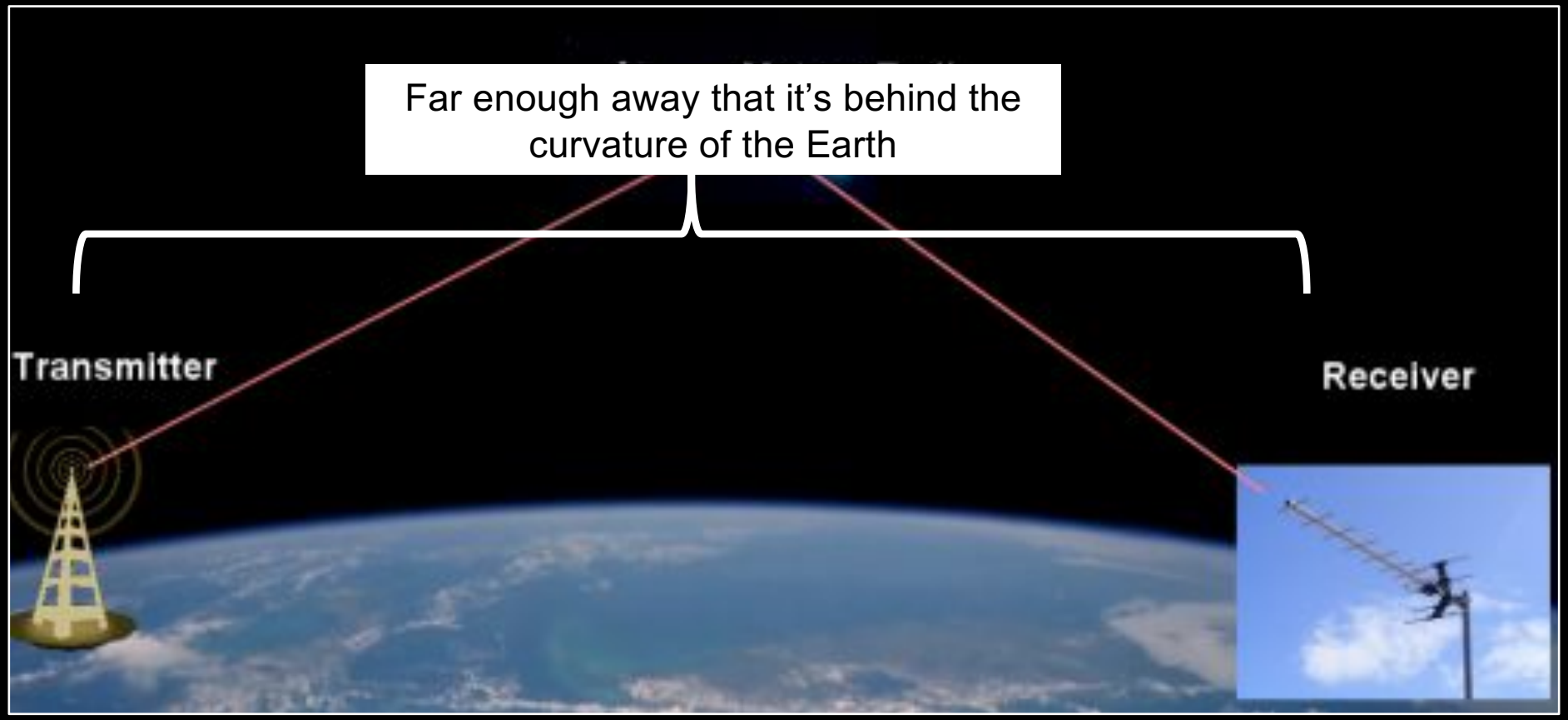
Inflight Meteors



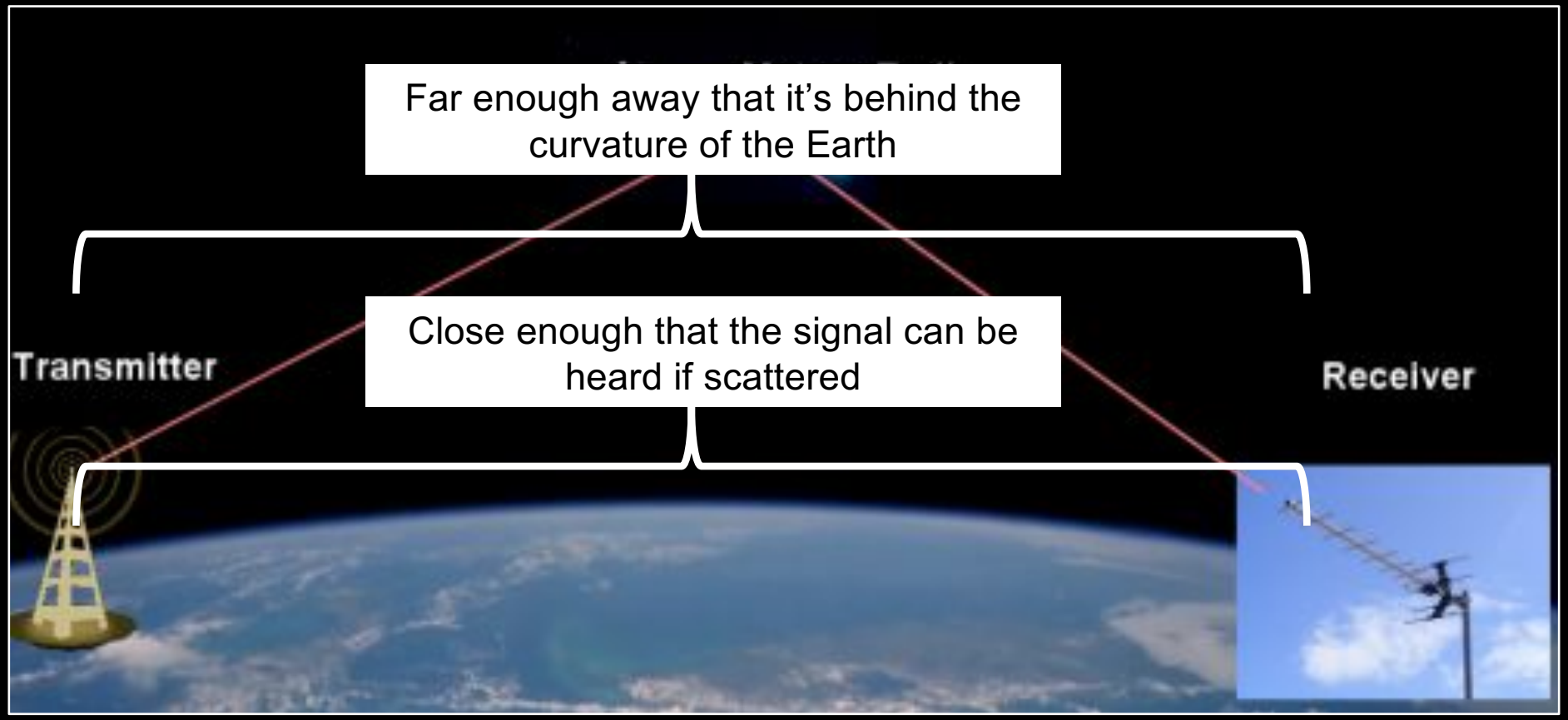
Meteor Detection – Inflight Meteors



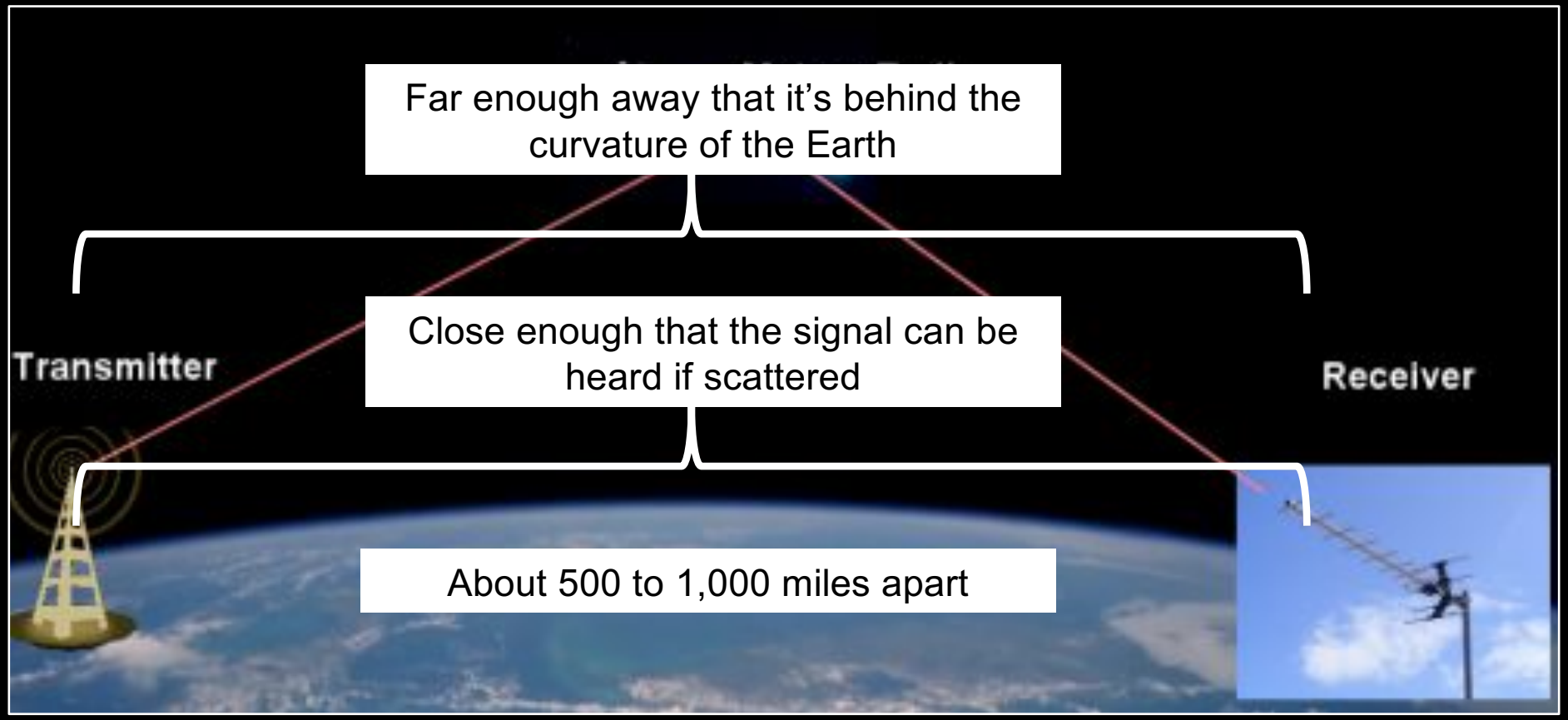
Meteor Detection – Inflight Meteors



Meteor Detection – Inflight Meteors



Meteor Detection – Inflight Meteors




WASHINGTON
Spokane
MONTANA
DAKOTA
Portland
Bozeman
Billings
Bend
SOUTH DAKOTA
OREGON
Boise
IDAHO
WYOMING
NEBRASKA
~710 miles
~635 miles
Salt Lake City
United States
NEVADA
Denver
Sacramento
UTAH
KANSAS
COLORADO
San Francisco
TV Station
CALIFORNIA
Las Vegas 54.31 MHz (Digital 2, Virtual 21)
Albuquerque
TV Stations
66.31 MHz (Digital 4, Virtual 22 & 63)
Los Angeles
ARIZONA
NEW MEXICO
Fort Worth
Phoenix
San Diego
Tucson
TEXAS

Meteor Detection – Inflight Meteors

History

1. First actually detected in WW2 by radar operators

2. Confirmed as meteors after the war



Meteor Detection – Inflight Meteors

Facts & Figures

1. Las Vegas at 54.31 MHz frequency

2. Las Vegas at ~18 foot wavelength

3. Dipole antenna at ½ wavelength (i.e., 9'), and 9' high; horizontal polarization

4. Software Defined Radio, SDR



Meteor Detection – Inflight Meteors

Facts & Figures

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2. Las Vegas at ~18 foot wavelength

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Meteor Detection – Inflight Meteors



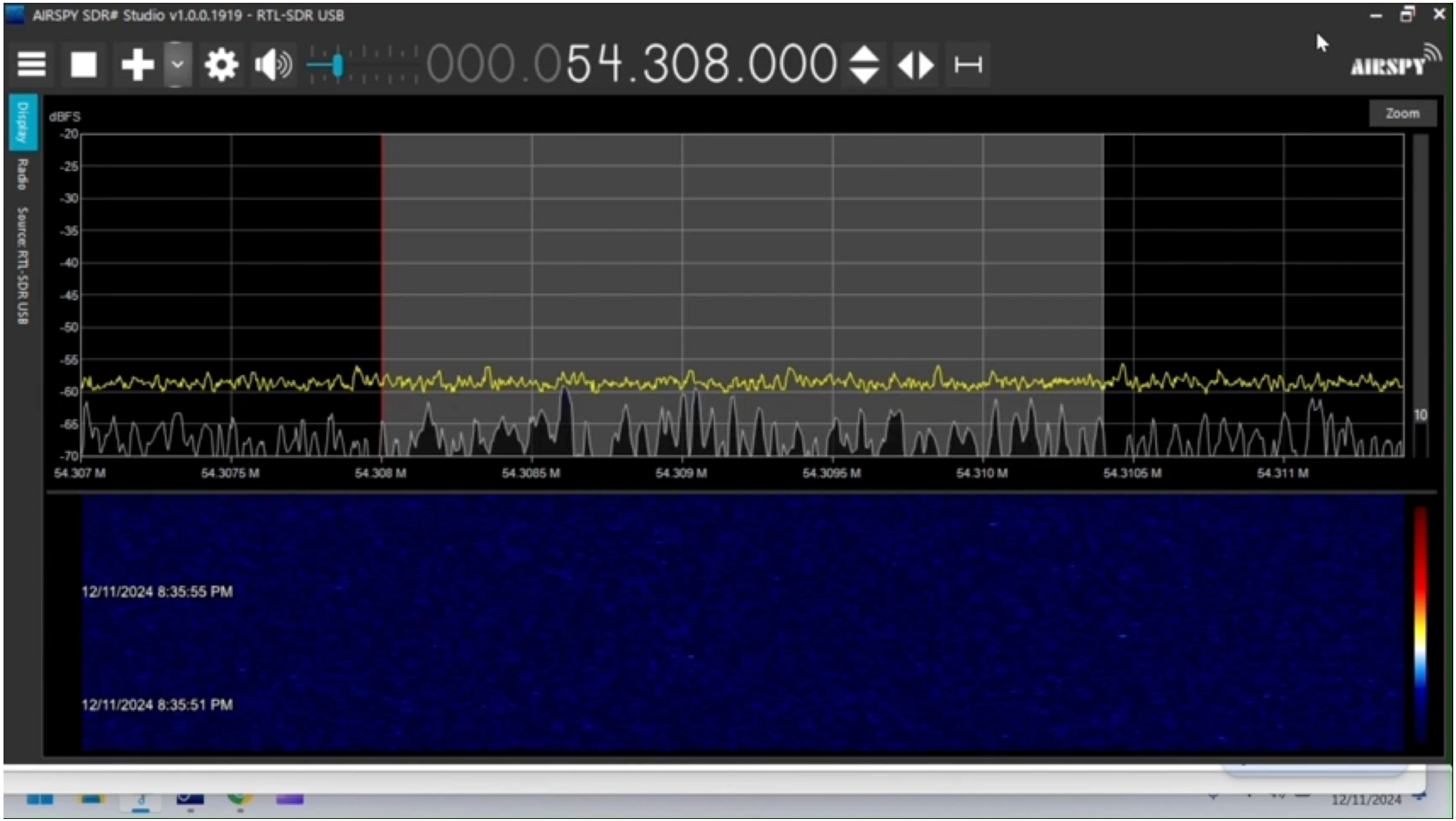


Meteor Detection – Inflight Meteors

Software Defined Radio - Software



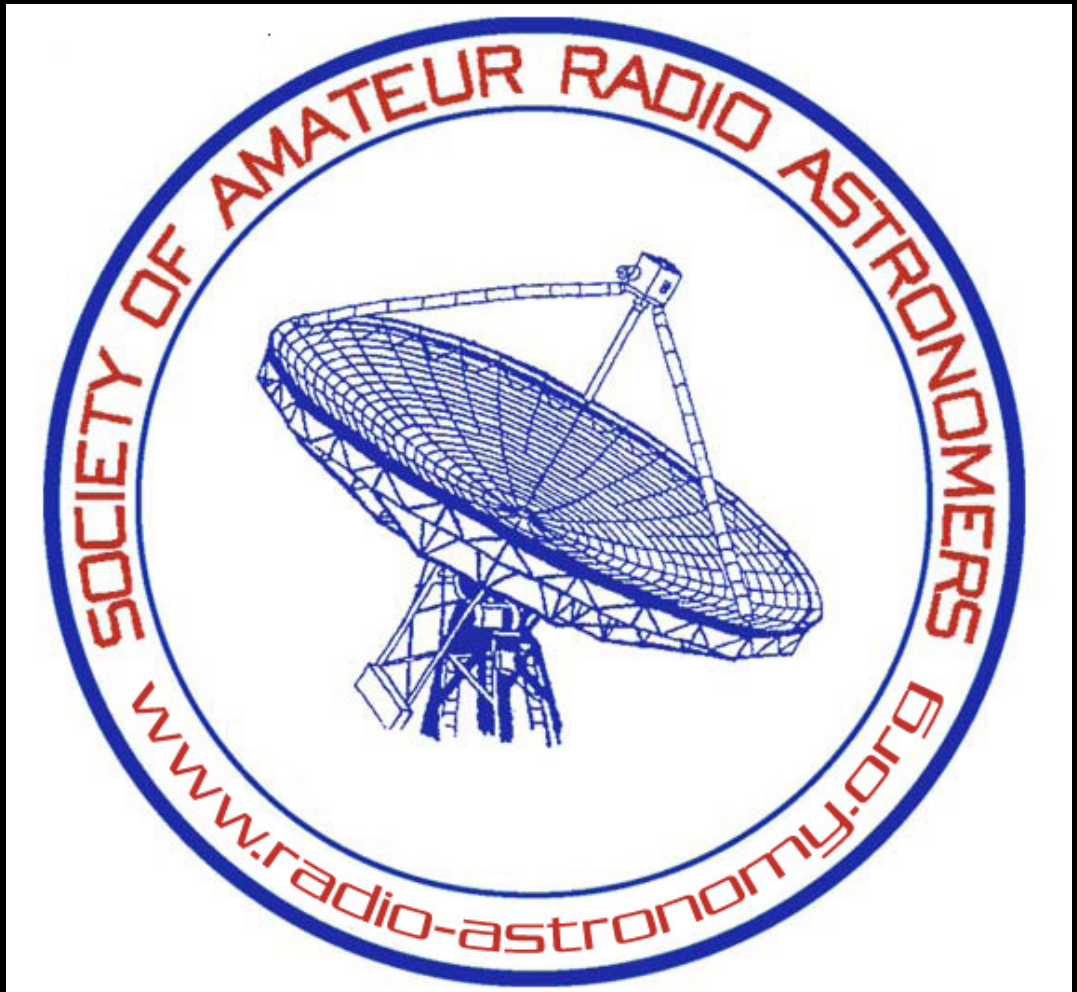
Software Defined Radio - Dongle


AIRSPY SDR# Studio v1.0.0.1919 - RTL-SDR USB
000.054.308.000
AIRSPY
Display
Radio
Source: RTL-SDR USB
dBFS
Zoom
-20
-25
-30
-35
-40
-45
-50
-55
-60
-65
-70
10
54.307 M
54.3075 M
54.308 M
54.3085 M
54.309 M
54.3095 M
54.310 M
54.3105 M
54.311 M
12/11/2024 8:35:55 PM
12/11/2024 8:35:51 PM
12/11/2024

Rose City Astronomers



Society of Amateur Radio Astronomers



Astronomical League



Thank You